Basic Coordinate Geometry

Exam code: 9MA0

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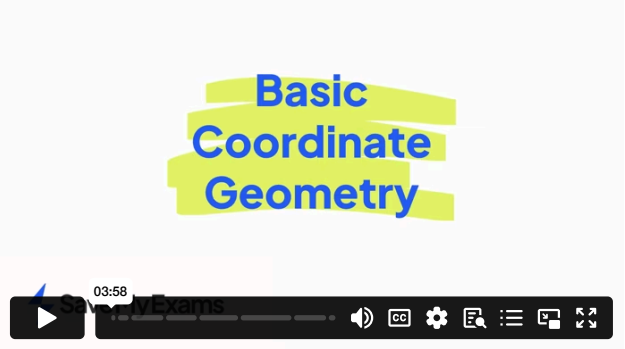

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Basic coordinate geometry

What is basic coordinate geometry?

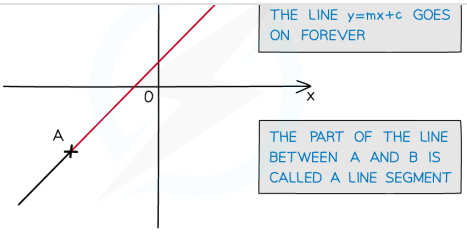
- Basic coordinate geometry refers to working with points, lines and shapes on the coordinate axis
- **Cartesian** coordinates are simply the **x** -**y** coordinate system
- Using coordinates you can
  - Calculate the distance between two points (length of a line)
  - Divide lines in m:n ratio,
  - Find the mid-point of a line
  - Calculate the area of a triangle (or other shapes)
  - Find missing coordinates using any of the above
- Most of these involve working with straight line graphs
- **Straight line graphs** are of the form **y = mx + c**
  - **m** is the gradient
  - **c** is the **y**-axis intercept
  - Part of a straight line is called a **line segment**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



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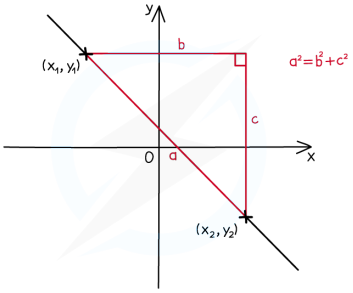
How do I find the length of a line segment?

- By finding the distance, **d**, between two points ($\mathbf{x_1}$, $\mathbf{y_1}$) and ($\mathbf{x_2}$, $\mathbf{y_2}$)
- **Pythagoras' Theorem** ($\mathbf{a^2 = b^2 + c^2}$) is used to find the length of a line



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$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

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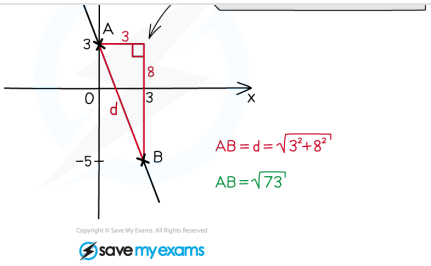
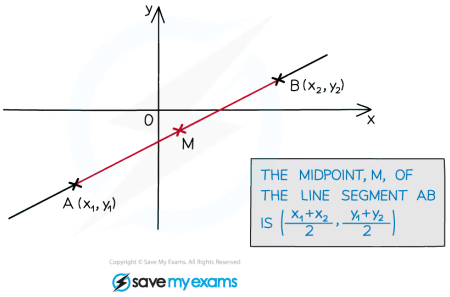


How do I find the midpoint of a line segment?



- M is often used as the midpoint between two points ($\mathbf{x}_1$, $\mathbf{y}_1$) and ($\mathbf{x}_2$, $\mathbf{y}_2$)
- It is the average of both the **x** and **y** coordinates

Examiner Tips and Tricks

- Work with the **square** of a distance for as long as possible as this avoids early rounding errors and surds.
- Only square root when forced to or for a final answer, and use the ANS button (and other memory features) on your calculator.

How do I find the gradient of a line segment?

- There are lots of ways to find the gradient, *m*, of a line segment
- Use the **two end points** to find the change in y divided by change in x

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Worked Example

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